

Nicholas C. Lopes

Department of Mechanical Engineering
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Education

Doctor of Philosophy in Mechanical Engineering Aug. 2023 - May 2026
Embry-Riddle Aeronautical University, Daytona Beach, FL
[Dissertation](#)
Advisor: Sandra K.S. Boetcher
Awards: 2025 Outstanding Doctoral Student in Mechanical Engineering

Master of Science in Mechanical Engineering Jan. 2021 - May 2023
Embry-Riddle Aeronautical University, Daytona Beach, FL
[Thesis](#)
Advisors: Sandra K.S. Boetcher and Mark A. Ricklick

Bachelor of Science in Aerospace Engineering Aug. 2017 - Dec. 2020
Embry-Riddle Aeronautical University, Daytona Beach, FL
Cum Laude

Research and Industry Experience

Oak Ridge National Laboratory (ORNL) Oak Ridge, TN (remote)
CFD Consultant — Multifunctional Equipment Integration Jan. 2026 - Jun. 2026

Oak Ridge National Laboratory (ORNL) Oak Ridge, TN (remote)
Energy Storage Research Intern — Multifunctional Equipment Integration Jan. 2025 - Dec. 2025

German Aerospace Center (DLR) Cologne, GER
Combustion Research Intern — Institute of Propulsion Technology Jun. 2024 - Aug. 2024

Pratt & Whitney West Palm Beach, FL
Aero-Thermal Engineer Intern — AeroThermal Fluids Group Jun. 2022 - Aug. 2022

Teaching Experience

Graduate Teaching Assistant
Department of Physical Science
Embry-Riddle Aeronautical University, Daytona Beach, FL
CHM 110L – General Chemistry I Laboratory (Spring 2021 - Fall 2024)

Guest Lecturer
Department of Mechanical Engineering
Embry-Riddle Aeronautical University, Daytona Beach, FL
ME 511 – Computational Heat Transfer (Spring 2024)
ME 409 – Vehicle Aerodynamics (Spring 2022, Fall 2022)

Journal Articles

- [J1] **N.C. Lopes**, J. Rendall, K Nawaz, S.K.S. Boetcher, J. Sun, Numerical analysis of embedded phase change materials in residential storage-type water heaters, *International Communications in Heat and Mass Transfer*, in preparation
- [J2] M.A. Messenger, **N.C. Lopes**, S.K.S. Boetcher, Characterization of Triply Periodic Minimal Surface Heat Exchangers with Latent Thermal Storage Capability, *International Journal of Heat and Mass Transfer*, submitted
- [J3] Y. Chao, **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, Criterion Sensitivity in the Determination of Thermal Entrance Length for Supercritical CO₂ Internal Flow, *Journal of Thermophysics and Heat Transfer*, submitted
- [J4] **N.C. Lopes**, C.M. Bochenek, M.A. Ricklick, S.K.S. Boetcher, Effects of cross-sectional shape on the cooling of supercritical carbon dioxide, *Case Studies in Thermal Engineering*, under revision
- [J5] C.J. Troxler, **N.C. Lopes**, A.J. Heiles, S.K.S. Boetcher, Reassessing widely adopted enthalpy–porosity modeling approaches for the numerical analysis of solid–liquid phase change, *Journal of Energy Storage*, under revision
- [J6] **N.C. Lopes**, Zachary R. Leonard, M.A. Ricklick, S.K.S. Boetcher, B. Janus, T. Behrendt, and R. Becker, Numerical investigation of gas turbine combustor cooling performance with Bézier effusion holes, *Journal of Thermophysics and Heat Transfer*, 2025.
[10.2514/1.T7282](#)
- [J7] **N.C. Lopes**, Y. Chao, M.A. Ricklick, S.K.S. Boetcher, Inlet condition effects on sCO₂ thermal behavior in water-cooled heat exchangers, *Journal of Thermophysics and Heat Transfer*, 2025.
[10.2514/1.T7211](#)
- [J8] **N.C. Lopes**, V. Dasarla, Y. Chao, P. Shekhar, S.K.S. Boetcher, M.A. Ricklick, Comparison of predictive tools for supercritical CO₂ horizontal cooling and the impact of artificial neural network input-output parameter selection, *Applied Thermal Engineering*, 2025.
[10.1016/j.applthermaleng.2024.125360](#)
- [J9] **N.C. Lopes**, Y. Chao, M.A. Ricklick, S.K.S. Boetcher, Influence of Thermal Boundary Conditions on Local sCO₂ Cooling Heat Transfer: a Case Study, *International Journal of Heat and Fluid Flow*, 2024.
[10.1016/j.ijheatfluidflow.2024.109310](#)
- [J10] Y. Chao, **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, Impact of adiabatic entrance length on numerical simulations of supercritical CO₂ heat transfer in horizontal circular tubes, *Numerical Heat Transfer, Part A: Applications*, 2023.
[10.1080/10407782.2023.2287534](#)
- [J11] **N.C. Lopes**, Y. Chao, V. Dasarla, N.P. Sullivan, M.A. Ricklick, S.K.S. Boetcher, Comprehensive Review of Supercritical CO₂ Correlations for Straight Tubes: A Historical Perspective, *Journal of Heat Transfer*, 2022.
[10.1115/1.4055345](#)
- [J12] Y. Chao, **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, Effect of the Heat Transfer Coefficient Reference Temperatures on Validating Numerical Models of Supercritical CO₂, *Journal of Verification, Validation and Uncertainty Quantification*, 2021.
[10.1115/1.4051637](#)

Conference Articles

- [C1] C.M. Bochenek, **N.C. Lopes**, M.A. Ricklick, and S.K.S. Boetcher, Investigating the effects of Prandtl’s secondary flows on sCO₂ heat transfer using Reynolds Stress Models, Proceedings of the 18th International Heat Transfer Conference, Rio de Janeiro, Brazil 2026. accepted

- [C2] C.M. Bochenek, **N.C. Lopes**, M.A. Ricklick, and S.K.S. Boetcher, Effects of Prandtl's secondary flows of the second kind on sCO₂ heat transfer, Paper no. 27, Proceedings of the 9th International sCO₂ Energy Technologies Symposium, Pittsburgh, PA, USA 2026.
[Symposium proceedings](#)
- [C3] K.J. Walsh, **N.C. Lopes**, C.M. Bochenek, M.A. Ricklick, S.K.S. Boetcher, Adiabatic entrance length selection for supercritical CO₂ flows in tubes with various hydraulic diameters, Proceedings of the AIAA SciTech Forum, Orlando, FL, USA 2026.
[10.2514/6.2026-2271](#)
- [C4] D.T. Hardy **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, Impact of Surface Roughness on Supercritical Carbon Dioxide Within a Gas-Cooler, Proceedings of the ASME International Mechanical Engineering Congress & Exposition, Memphis, TN, USA 2025.
[10.1115/IMECE2025-165151](#)
- [C5] **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, B. Janus, T. Behrendt, and R. Becker, Numerical investigation of gas turbine combustor cooling performance with Bézier effusion holes, Proceedings of the AIAA SciTech Forum, Orlando, FL, USA 2025.
[10.2514/6.2025-2503](#)
- [C6] **N.C. Lopes**, Y. Chao, M.A. Ricklick, and S.K.S. Boetcher, Impact of inlet conditions on local sCO₂ heat exchanger performance, Proceedings of the AIAA SciTech Forum, Orlando, FL, USA 2025.
[10.2514/6.2025-2846](#)
- [C7] Y. Chao, **N.C. Lopes**, S.K.S. Boetcher, and M.A. Ricklick, Analysis of Thermal Entrance Length in Turbulent Supercritical CO₂ Flow through a Horizontal Circular Tube, Proceedings of the AIAA SciTech Forum, Orlando, FL, USA 2025.
[10.2514/6.2025-2844](#)
- [C8] D. Velasco, J. Sauerbrun, **N.C. Lopes**, Y. Chao, S.K.S. Boetcher, and M.A. Ricklick, Benchmarking of a sCO₂ Heat Exchanger Flow Loop, Proceedings of the AIAA SciTech Forum, Orlando, FL, USA 2024.
[10.2514/6.2024-2294](#)
- [C9] Y. Chao, **N.C. Lopes**, S.K.S. Boetcher, and M.A. Ricklick, Numerical investigation of flow development and thermal boundary condition effects on local sCO₂ heat transfer trends in circular tubes, Proceedings of the AIAA SciTech Forum, National Harbor, MD, USA 2023.
[10.2514/6.2023-0604](#)
- [C10] V. Dasarla, Y. Chao, **N.C. Lopes**, M.A. Ricklick, P. Shekhar, and S.K.S. Boetcher, Impact of Data Representation on Artificial Neural Network Performance in sCO₂ Cooling Applications, Proceedings of the AIAA SciTech Forum, National Harbor, MD, USA 2023.
[10.2514/6.2023-0390](#)
- [C11] Y. Chao, **N.C. Lopes**, M.A. Ricklick, and S.K.S. Boetcher, Hydraulic development length and boundary condition effects on local sCO₂ heat transfer coefficients, Paper no. 182, Proceedings of the 7th International Supercritical CO₂ Power Cycles Symposium, San Antonio, TX, USA 2022.
[Symposium proceedings](#)

Invited Posters and Presentations

- [P1] C.M. Bochenek, **N.C. Lopes**, M.A. Ricklick, and S.K.S. Boetcher, Effects of Prandtl's secondary flows of the second kind on sCO₂ heat transfer, Poster no. 27, 9th International sCO₂ Energy Technologies Symposium, Pittsburgh, PA, USA 2026. (Poster)
- [P2] K.J. Walsh, **N.C. Lopes**, C.M. Bochenek, M.A. Ricklick, S.K.S. Boetcher, Adiabatic entrance length selection for supercritical CO₂ flows in tubes with various hydraulic diameters, AIAA SciTech Forum, Orlando, FL, USA 2026. (Presentation)

- [P3] C.M. Bochenek, **N.C. Lopes**, M.A. Ricklick, and S.K.S. Boetcher, Investigating the effects of secondary flows on supercritical CO₂ heat transfer, ERAU Student Research Symposium 2025, Daytona Beach, FL, USA 2025. (Poster)
- [P4] **N.C. Lopes**, K.J. Walsh, Y. Chao, M.A. Ricklick, S.K.S. Boetcher, Evaluating development length for supercritical CO₂ tube flows, Summer Heat Transfer Conference, Westminster, CO, USA 2025. (Poster)
- [P5] **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, B. Janus, T. Behrendt, and R. Becker, Numerical investigation of gas turbine combustor cooling performance with Bézier effusion holes, AIAA SciTech Forum, Orlando, FL, USA 2025. (Presentation)
- [P6] **N.C. Lopes**, Y. Chao, M.A. Ricklick, and S.K.S. Boetcher, Impact of inlet conditions on local sCO₂ heat exchanger performance, AIAA SciTech Forum, Orlando, FL, USA 2025. (Presentation)
- [P7] Y. Chao, **N.C. Lopes**, S.K.S. Boetcher, and M.A. Ricklick, Analysis of Thermal Entrance Length in Turbulent Supercritical CO₂ Flow through a Horizontal Circular Tube, AIAA SciTech Forum, Orlando, FL, USA 2025. (Presentation)
- [P8] Y. Chao, K.J. Walsh **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, Transcritical CO₂ flow through a horizontal circular tube: A thermal entrance length analysis, Proceedings of the Florida Fluids Symposium III, Daytona Beach, FL, USA 2024. (Presentation)
- [P9] K.J. Walsh, Y. Chao, **N.C. Lopes**, M.A. Ricklick, S.K.S. Boetcher, Impact of diameter and operating conditions on heat transfer of supercritical CO₂ in horizontal circular tubes, Proceedings of the Florida Fluids Symposium III, Daytona Beach, FL, USA 2024. (Presentation)
- [P10] **N.C. Lopes**, V. Dasarla, Y. Chao, M.A. Ricklick, P. Shekhar, S.K.S. Boetcher, Impact of Data Representation on Artificial Neural Network Performance in sCO₂ Cooling Applications, AIAA SciTech Forum, National Harbor, MD, USA 2023. (Presentation)
- [P11] Y. Chao, **N.C. Lopes**, S.K.S. Boetcher, and M.A. Ricklick, Numerical investigation of flow development and thermal boundary condition effects on local sCO₂ heat transfer trends in circular tubes, AIAA SciTech Forum, National Harbor, MD, USA 2023. (Presentation)
- [P12] Y. Chao, **N.C. Lopes**, M.A. Ricklick, and S.K.S. Boetcher, Hydraulic development length and boundary condition effects on local sCO₂ heat transfer coefficients, Poster no. 182, 7th International Supercritical CO₂ Power Cycles Symposium, San Antonio, TX, USA 2022. (Poster)

Funded Grants and Proposals

- [G1] DOE: UT-Batelle, LLC Fixed Price Subcontract CW72370
Risk Assessment of HVAC Equipment Using A3 Refrigerants
 Role: Graduate Research Assistant (PI: Sandra K.S. Boetcher)
 Sponsor: UT-Batelle, LLC c/o Oak Ridge National Laboratory
 Dates: 1/1/26 - 6/30/26 Funding: \$28,000
- [G2] DOE: UT-Batelle, LLC Fixed Price Subcontract CW62734
Natural Convection Study of Bio-Based Phase-Change Materials for Thermal Energy Storage in Heat Pump Water Heaters
 Role: Graduate Research Assistant (PI: Sandra K.S. Boetcher)
 Sponsor: UT-Batelle, LLC c/o Oak Ridge National Laboratory
 Dates: 1/1/25 - 12/31/25 Funding: \$43,000
- [G3] ERAU: Philanthropy Council Grant
Enabling Environmentally Friendly Refrigeration through Supercritical Fluid Experiments

Role: Graduate Student (PIs: Mark A. Ricklick, Sandra K.S. Boetcher)

Sponsor: ERAU

Dates: 9/1/22 - 6/30/23

Funding: \$5,000

[G4] ASHRAE: Senior Undergraduate Equipment Grant

Green HVAC/R with Supercritical Carbon Dioxide Refrigerant

Role: Graduate Student (PIs: Sandra K.S. Boetcher, Mark A. Ricklick)

Sponsor: ASHRAE

Dates: 9/1/22 - 8/31/23

Funding: \$5,000

Professional Affiliations

Current

ASME American Society of Mechanical Engineers

Past

ASHRAE American Society of Heating, Refrigerating, and Air-Conditioning Engineers

AIAA American Institute for Aeronautics and Astronautics

Scholarly Reviews

ASME: Open Journal of Engineering
 Journal of Thermal Science and Engineering Applications
 Journal of Heat and Mass Transfer

Elsevier: International Communications in Heat and Mass Transfer
 International Journal of Heat and Fluid Flow